Calculating Limits Numerically

Let $f(x) = 2x^3 - 4x + 1$ and let $g(x) = \dfrac{e^x - 1}{x}$.

(a) Use Python code to evaluate the function f at appropriate values to calculate $\lim_{x \to 3} f(x)$. **Hint:** Evaluate f at values closer and closer to 3 and see what value the outputs get close to.

(b) Use Python code to evaluate the function g at appropriate values to calculate $\lim_{x \to 0} g(x)$.

(c) Could the limits in parts (a) and (b) be calculated by plugging in the value where the limit is taken into the function? Explain. What does this mean in terms of continuity?

Average Rate of Change

The formula for the average rate of change of a function f between two values a and b is given by

$$\text{Average Rate of Change } = \frac{f(b) - f(a)}{b - a}.$$

Create a Python function that takes in a mathematical function, e.g., $f(x) = 3x^2$ and two numbers a and b and returns the average rate of change of the function between a and b.

Average Rate of Change to Instantaneous Rate of Change

A baseball is dropped from a tall cliff. Neglecting air resistance, the distance traveled by the baseball in meters after t seconds is given by the function

$$f(t) = 4.9t^2.$$

(a) Find the average speed (average rate of change of distance) of the baseball between 5 and 6 seconds.

(b) Find the average speed (average rate of change of distance) of the baseball between 5 and 5.5 seconds.

(c) Find the average speed (average rate of change of distance) of the baseball between 5 and 5.1 seconds.

(d) What is the instantaneous speed of the baseball at $t = 5$ seconds? **Hint:** You might want to do a few more calculations similar to those done in parts (a) - (c).

(e) Find the derivative of f. **Hint:** The derivative of a quadratic function $f(x) = ax^2 + bx + c$ is given by $f'(x) = 2ax + b$.

(f) Evaluate the derivative of f at $t = 5$. How does this value compare to the value found in part (d)? Explain what is happening.

Calculating and Interpreting Partial Derivatives

A multivariate linear model is trained to predict the selling price of a particular used car model. The linear model predicts the selling price (P) in US dollars, using its current condition (C) on a scale of $1 - 10$ and the age of the car in years (Y). The model is given by

$$P = 16,000 + 2,400C - 1,800Y.$$

(a) What is this model's predicted selling price of a 5-year old car with a condition rating 8?

(b) Find $\dfrac{\partial P}{\partial C}$ and interpret this value.

(c) Find $\dfrac{\partial P}{\partial Y}$ and interpret this value.